Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

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Osimertinib in Untreated *EGFR*-Mutated Advanced Non–Small-Cell Lung Cancer

SUPPLEMENTARY APPENDIX

This appendix is provided by the authors to give readers additional information about their work.

Supplement to: Osimertinib in Untreated *EGFR*-Mutated Advanced Non–Small-Cell Lung Cancer; Soria JC, Ohe Y, Vansteenkiste J et al. 2017

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SUPPLEMENTARY METHODS

Trial oversight

All authors signed a confidentiality agreement with the sponsor. An agreement was in place between the trial sponsor and the authors, which established the authors' rights to publish the trial and access the data. Responsibility for opinions, conclusion, and interpretation of the data lies with the authors.

Trial design

Individual sites were required to select either gefitinib (250 mg, once daily) or erlotinib (150 mg, once daily) as the sole comparator prior to study initiation, except for the US, where all sites used erlotinib.

Trial assessments

All patients underwent tumor imaging (CT or MRI) at baseline within 4 weeks before randomization. Subsequent assessments were every 6 weeks for the first 18 months then every 12 weeks until disease progression. Patients with known or suspected central nervous system (CNS) metastases at trial entry were mandated to have baseline brain scans and patients with confirmed CNS metastases received follow-up brain scans. Any new lesions found in the CNS during the trial were identified as new lesions.

Definitions of secondary endpoints

Overall, where the analysis models are stratified according to the randomization stratification factors, race and mutation status, the strata obtained at randomization were used, not the values recorded in the electronic case report form (eCRF).

Objective response rate was defined as the proportion of randomized patients with at least one visit response of complete response or partial response. Duration of response was defined as the time from the date of first documented response until the date of documented progression or death in the absence of disease progression. If a patient did not progress or die following a response, then the progression-free survival censoring time was used. Disease control rate was defined as the percentage of patients who had a best overall response of complete response, partial response or stable disease of ≥6 weeks, prior to any disease progression event. Depth of response was defined as the relative change in the sum of the longest diameters of RECIST target lesions at the nadir in the absence of new lesions, or progression of non-target lesions compared with baseline. The depth of response was the largest decrease from baseline (or the smallest increase from baseline in the absence of a

reduction) and included all assessments prior to progression or initiation of further anticancer therapy. Overall survival was defined as the time from randomization until death due to any cause. Patients not known to have died at the time of analysis were censored based on the last recorded date on which the patient was known to be alive.

Definitions of exploratory endpoints

Time to first subsequent therapy was defined as time from the date of randomization to the earliest start date of subsequent anti-cancer therapy following trial drug discontinuation, or death. Any patient not known to have received a first subsequent therapy, or to have died at the time of the analysis, was to be censored at the last date the patient was known not to have received a first subsequent therapy (i.e., the last follow-up visit where this was confirmed).

PFS2 was defined as the time from randomization to second progression, i.e., time from date of randomization to the earliest of the progression events subsequent to that used for the primary efficacy variable of PFS or date of death after starting subsequent anti-cancer treatment. The assessment of the impact of baseline variables was the same as for PFS, provided there were sufficient events available for analysis. If a patient died without any progression event, the patient's PFS and PFS2 event dates were equivalent. If a patient died after a primary PFS event but prior to the initiation of a subsequent anti-cancer therapy, the date of death was considered to be the PFS2 event. Patients alive and for whom no second disease progression was observed were censored at the last time they were known to be alive and without second disease progression, i.e. at the last progression/disease assessment date (or Day 1 if no post-baseline RECIST data were available) if the patient did not have a second progression or died.

Time to second subsequent therapy was defined as time from the date of randomization to the earliest start date of the second subsequent anti-cancer therapy following trial drug discontinuation, or death. Any patient not known to have received a second subsequent therapy, or to have died at the time of the analysis, was to be censored at the last date the patient was known not to have received a second subsequent therapy (i.e., the last follow-up visit where this was confirmed).

Safety assessments

Safety assessments included adverse events graded by National Cancer Institute Common Terminology Criteria for Adverse Events (CTCAE) version 4.0; clinical chemistry, hematology and urinalysis; vital signs, physical examination, body weight; digital

electrocardiogram; left ventricular ejection fraction; World Health Organization Performance Status; and ophthalmologic assessment.

Adverse events and serious adverse events were listed individually by patient by MedDRA preferred terms and CTCAE grade. Specified adverse events of special interest were provided as grouped terms. For patients with multiple preferred term level events within a specific grouped term adverse event, the maximum CTCAE grade across those events was counted. Any adverse events occurring within 28 days of the last dose of trial treatment or the start of crossover treatment were included in the adverse event summaries. Safety data were summarized but not formally analyzed.

To identify any clinically significant risks to patients, investigator-reported adverse events for cardiac effects (QT or cardiac failure) were assessed together with changes in QTcF and cardiac contractility laboratory values observed during treatment with osimertinib or standard EGFR-TKI. Electrocardiogram (ECG) recordings (performed at screening and then throughout treatment) were used to assess QT prolongation. Assessment of left ventricular ejection fraction (LVEF) was performed by echocardiography/multi-gated acquisition (MUGA) scans performed at screening, every 12 weeks relative to the first dose, and at treatment discontinuation.

Dose modifications

Where dose modifications were required to manage toxicity, patients were required to undergo dose interruption prior to considering a dose reduction. Dosing was interrupted, and supportive therapy administered as required in accordance with local practice/guidelines, in patients experiencing a CTCAE grade 3 or higher adverse event and/or unacceptable toxicity (any grade), not attributable to the disease under investigation, and considered by the investigator to be specifically associated with the trial drug. If the toxicity resolved or reverted to CTCAE grade ≤1 within 2 weeks of onset, the trial drug was restarted at the same dose, or reduced dose using the dose reduction levels described below. No dose reduction for gefitinib was possible; thus, the reduced dose for gefitinib was the same as the starting dose because the 250 mg tablets are the lowest dose available. If restarting at the same dose level, patients were closely monitored for 3 days following the restart of treatment. If, within 3 days, there was a recurrence of the same toxicity, a dose reduction was considered at the investigator's discretion. If the toxicity did not resolve to CTCAE grade ≤1 after 2 weeks, then the patient was withdrawn from the trial treatment and observed until resolution of the toxicity. There was no individual modification to the treatment schedule in response to toxicity, only potential dose reduction or dose interruption. If an adverse event subsequently required a dose interruption, the trial drug was restarted at the same dose, on

resolution/improvement of the adverse event at the discretion of the investigator as described above.

Dose reduction levels

	Osimertinib	Gefitinib	Erlotinib
Starting dose	80 mg osimertinib/comparator matching placebo	250 mg gefitinib/osimertinib matching placebo	150 mg erlotinib/osimertinib matching placebo
Reduced dose	40 mg osimertinib/comparator matching placebo	NA*	100 mg erlotinib/osimertinib matching placebo

NA, not applicable
* No dose reduction was possible for gefitinib.

Statistical analysis

Sample size

It was determined that 359 events of progression or death from 530 randomized patients would be required to achieve 90% power to detect a hazard ratio of 0.71 (representing an improvement in median progression-free survival from 10 months to 14.1 months assuming exponential data distribution and proportional hazards) at a two-sided alpha-level of 5%. The minimum critical hazard ratio was defined as 0.81 (improvement in median progression-free survival from 10 to 12 months).

For overall survival, it was determined that the target sample size would provide approximately 72% power to demonstrate a hazard ratio <0.75 (improvement in overall survival from 25 to 33.3 months) with a two-sided 5% significance level. In the event that the 4-month improvement projected for progression-free survival was maintained for overall survival, this would provide approximately 27% power to detect a hazard ratio of 0.86 (median overall survival 25 months for standard EGFR-TKI and 29 months for osimertinib) with a two-sided 5% significance level.

Analysis sets

The full analysis set included all globally randomized patients, including patients who were randomized but did not subsequently receive treatment. All efficacy analyses were performed on the full analysis set and treatment groups were compared on the basis of randomized trial treatment, regardless of the treatment actually received.

The safety analysis set included all globally recruited patients who received at least one dose of trial treatment. Safety data were summarized using the safety analysis set, according to the treatment actually received, regardless of randomization group.

Multiple testing strategy

To control the type I error rate, a sequential multiple testing strategy was used. Progression-free survival (primary endpoint), overall survival, and CNS progression-free survival were tested in this sequential order. If any previous analysis in the sequence was not statistically significant, then significance testing of the subsequent endpoints would not be performed.

Analysis of progression-free survival

Investigator assessed progression-free survival was analyzed using a log-rank test stratified by race (Asian vs. non-Asian) and mutation type (exon 19 deletion vs. L858R), applying the Breslow approach to handle tied events.

The hazard ratio (HR) and confidence interval (CI) were obtained directly from the U and V statistics as follows:

$HR = \exp(U/V)$

95% CI for HR = $(\exp[U/V - 1.96/\sqrt{V}], \exp[U/V + 1.96\sqrt{V}])$

Where $U = \sum_k U_k = \sum_k \sum_i (d_{1ki} - e_{1ki})$ is the stratified log-rank test statistic (with d_{1ki} and e_{1ki} the observed and expected events in group 1, stratum k) and $\sqrt{V} = \sqrt{\sum_k V_k}$ is the standard deviation obtained from the LIFETEST procedure with a STRATA term for the stratification variable. The assumption of proportionality was assessed by examining plots of complementary $\log\text{-}\log_{\text{event times}}$ versus $\log_{\text{time}}$.

A Kaplan-Meier plot of progression-free survival by treatment group was generated. The total number of events, median progression-free survival and the percentage of patients who were progression-free at 6, 12, 18 and 24 months were summarized. The progression status at the time of the progression-free survival analysis were summarized, including the number and percentage of patients who progressed due to RECIST progression or death, or did not progress due to being progression-free or lost to follow up.

The possibility of bias in assessment and measurement of progression-free survival by investigators was assessed by comparing the HRs derived from investigator review with the HR derived using the blinded independent central review (BICR) assessment of disease progression by RECIST. In order to assess possible evaluation-time bias that could occur if scans were not performed at the protocol-scheduled time points, the midpoint between the

time of progression and the previous evaluable RECIST assessment was analyzed using a stratified log rank test, as described for the primary analysis of progression-free survival. For patients who died in the absence of progression, the date of death was used to derive the progression-free survival time used in the analysis. The possibility of bias resulting from the rate and nature of censoring (attrition bias) was assessed by repeating the primary progression-free survival analysis, except that the actual progression-free survival event times, rather than the censored times, of patients who progressed or died in the absence of progression immediately following two or more non-evaluable tumor assessments were included. In addition, patients who had subsequent therapy prior to progression or death were censored at their last evaluable assessment prior to taking the subsequent therapy. This analysis was supported by a Kaplan-Meier analysis of the time to censoring using the progression-free survival data from the primary analysis, where the censoring indicator of the primary progression-free survival analysis was reversed, to assess the number of patients being followed over time. The possibility of bias due to the proximity of the median progression-free survival in the osimertinib group to the change in assessment schedule from every 6 to every 12 weeks at month 18 was assessed by moving all progression events backwards 6 weeks to artificially create a 6 week visit. Progression-free survival events of death and censored patients remained per the primary analysis.

For the following pre-defined subgroup analyses, HRs and 95% CIs were calculated using a Cox proportional hazards model containing a term for treatment, the subgroup covariate of interest and the treatment by subgroup interaction term:

- Sex (male vs. female)
- Race (Asian vs. non-Asian)
- Age at screening (<65 years vs. ≥65 years)
- CNS metastases status at entry (yes vs. no)
- Smoking history
- Baseline WHO Performance Status
- Pre-treatment T790M status
- EGFR mutation (exon 19 deletion vs. L858R)
- EGFR mutation-positive by ctDNA
- Centrally confirmed EGFR mutation

No adjustment to the significance level was made since these subgroup analyses were only designed to be supportive of the primary analysis of progression-free survival.

Analysis of CNS metastases

The proportion of patients who reported CNS metastases during the trial, whether solely in the CNS or at the same time as non-CNS lesions, were summarized by treatment group.

Analysis of overall survival

Overall survival was assessed at the time of the primary analysis of progression-free survival and will be assessed again at approximately 60% maturity. As two analyses are planned, the Lan DeMets approach that approximates the O'Brien and Fleming spending function was used to maintain an overall 2-sided 5% type I error across this interim analysis of overall survival and the planned analysis at 60% maturity. The significance level for the analyses was calculated using the information fraction for each analysis, defined as the number of overall survival events at the analysis time point divided by the total number of events at the final analysis time point.

Overall survival was analyzed using the same methodology and model as for the analysis of progression-free survival.

Analysis of overall response rate and disease control rate

Overall response rate and disease control rate were analyzed using a logistic regression stratified by race (Asian vs. non-Asian) and mutation type (exon 19 deletion vs. L858R).

Analysis of duration of response

Investigator assessed duration of response was analyzed using the Kaplan-Meier method, stratified by race (Asian vs. non-Asian) and mutation type (exon 19 deletion vs. L858R).

Analysis of depth of response

Depth of response was examined by the proportion of patients achieving >30%, >50% and >75% reduction in target lesion tumor size. The effect of osimertinib on best percentage change in tumor size was estimated from an analysis of covariance model with covariates for race (Asian vs. non-Asian) and mutation type (exon 19 deletion vs. L858R), baseline tumor size and time from baseline scan to randomization.

Rationale for amendments to the trial protocol

Following an amendment to the trial protocol on April 13, 2015 (amendment 1), patients randomized to receive standard EGFR-TKI were permitted to cross over to receive open-label osimertinib, provided central confirmation of objective disease progression (per RECIST version 1.1) by BICR was obtained while still blinded, there had been no intervening therapy following treatment discontinuation, and post-progression confirmation of T790M-positive mutation status was obtained. This amendment was based on emerging data from

the wider osimertinib development program that indicated a favorable risk benefit in T790M-positive patients who progressed following treatment with an early generation EGFR-TKI. The trial sponsor considered it appropriate that patients who were randomized to the standard EGFR-TKI treatment group had the option to receive osimertinib if they fulfilled the criteria outlined above.

In a second amendment to the FLAURA trial protocol, dated September 24, 2015 (amendment 2), the defined sample size for randomization was reduced from 650 to 530 patients. The decision for this change was based on positive data from the phase 1 AURA trial, showing that, at time of data cutoff (2 December 2014), 81% of those patients (n=60) treated with first-line osimertinib were alive and progression free at 12 months. As a result, the target magnitude of treatment difference between osimertinib and standard EGFR-TKI in FLAURA was increased, reducing the target hazard ratio for progression-free survival from 0.74 to 0.71. This translates into an approximate improvement in median progression-free survival from 10 months with standard EGFR-TKI to 14.1 months with osimertinib.

The trial protocol and statistical analysis plan are available with the full text of this article at NEJM.org.

SUPPLEMENTARY RESULTS

Patients

Per protocol, all patients were to receive trial treatment as first-line therapy. However, 3 (0.5%) patients (osimertinib: 2 [0.7%]; standard EGFR-TKI: 1 [0.4%]), all enrolled at the same US trial site, were not treatment-naïve for locally-advanced or metastatic NSCLC at trial entry, an important protocol deviation.

T790M mutation status at baseline

Five patients in the full analysis set were T790M positive based on tissue and/or ctDNA testing; 4 in the osimertinib group, 1 in the standard EGFR-TKI group.

Progression-free survival (PFS)

At the time of data cutoff, 136 patients (49%) in the osimertinib group, and 206 (74%) in the standard EGFR-TKI group, had experienced a progression event or death (overall maturity: 62%).

The sensitivity analysis of PFS according to blinded independent central review was consistent with the investigator analysis. At time of data cutoff, 137 patients (49%) in the osimertinib group, and 198 (71%) in the standard EGFR-TKI group, had experienced a RECIST progression event or death. The hazard ratio for disease progression or death favored osimertinib (0.45 [95% CI 0.36 to 0.57], P<0.001). Median PFS was 17.7 months (95% CI 15.1 to 21.4) with osimertinib and 9.7 months (95% CI 8.5 to 11.0) with standard EGFR-TKI (Figure S2).

The sensitivity analysis for attrition bias, using actual PFS event times rather than censored times, was consistent with the investigator analysis. One hundred and forty one patients (51%) in the osimertinib group, and 202 (73%) in the standard EGFR-TKI group, had experienced a RECIST progression event or death. The hazard ratio for disease progression or death was 0.47 (95% CI 0.37 to 0.58), P<0.001.

The sensitivity analysis to assess the impact of the change in visit schedule at 18 months showed the impact to be very minor, with the hazard ratio for disease progression or death, remaining unchanged from the primary analysis (0.46 [95% CI 0.37 to 0.58]; P<0.001).

Time to discontinuation of trial treatment or death

The median time to discontinuation of randomized treatment or death was longer in the osimertinib group (20.8 months) versus the standard EGFR-TKI group (11.5 months). The proportion of patients continuing treatment beyond RECIST progression was similar in the osimertinib and standard EGFR-TKI groups (osimertinib 67%, standard EGFR-TKI, 70%). The median duration of treatment beyond progression was also similar (8.1 weeks with osimertinib vs. 7.0 weeks with standard EGFR-TKI).

Subsequent therapy

Overall, 115 patients (41%) in the osimertinib group and 175 (63%) in the standard EGFR-TKI group started a first subsequent anticancer therapy or died. In the osimertinib group, 82 patients (29%) received a first subsequent anticancer therapy versus 129 patients (47%) in the standard EGFR-TKI group. An additional 33 patients (12%) in the osimertinib group and 46 (17%) in the standard EGFR-TKI group died in the absence of a first subsequent therapy.

Of those patients who discontinued randomized trial treatment, patients on osimertinib most commonly received chemotherapy (platinum or non-platinum based; 71% vs. 25% in the standard EGFR-TKI group) and patients on standard EGFR-TKI most commonly received EGFR-TKI-containing therapy (46% vs. 21% in the osimertinib group). There was a statistically significant delay in initiation of first subsequent anticancer therapy or death in the osimertinib group compared to the standard EGFR-TKI group (HR: 0.51 [95% CI 0.40 to 0.64], P<0.001). The median time to first subsequent therapy or death was 23.5 months (95% CI 22.0 to not calculable [NC]) on osimertinib and 13.8 months (95% CI 12.3 to 15.7) on standard EGFR-TKI.

Time from randomization to second progression after start of subsequent treatment (PFS2) data showed a clinically meaningful improvement for osimertinib, HR 0.58 (95% CI 0.44 to 0.78; P<0.001). At the time of analysis, 73 (26%) patients on osimertinib and 106 (38%) patients on SoC had second progression events after the start of subsequent therapy or died. The median PFS2 was NC (95% CI 23.7 to NC) and 20.0 months (95% CI 18.2 to NC) for osimertinib and standard EGFR-TKI groups respectively.

Overall, 74 (27%) osimertinib patients and 110 (40%) of the standard EGFR-TKI patients began a second subsequent therapy or died. In the osimertinib group, 24 patients (9%) received a second subsequent anticancer therapy versus 39 patients (14%) in the standard EGFR-TKI group and the remaining patients were those who died. There was a clinically meaningful improvement in the time to second subsequent therapy in the osimertinib group

compared to the standard EGFR-TKI group: HR 0.60 (95% CI 0.45 to 0.80; P<0.001). The median time to second subsequent therapy or death was not calculable for osimertinib (95% CI NC to NC) and was 25.9 months (95% CI 20.0 to NC) in the standard EGFR-TKI group.

Objective response rate (ORR) and duration of response (DoR) in patients with and without known/treated CNS metastases status at trial entry

In patients with known/treated CNS metastases at trial entry (CNS metastases status = yes) (osimertinib: n=53; standard EGFR-TKI: n=63), ORR (by investigator) was 75% (95% CI 62 to 86) in the osimertinib group and 86% (95% CI 75 to 93) in the standard EGFR-TKI group. The median DoR was 13.8 months (95% CI 10.8 to 20.2) in the osimertinib group and 8.3 months (95% CI 5.5 to 9.6) in the standard EGFR-TKI group. In patients without known/treated CNS metastases at trial entry (CNS metastases status = no), ORR (by investigator) was 81% (95% CI 75 to 86) in the osimertinib group and 73% (95% CI 66 to 79) in the standard EGFR-TKI group. The median DoR was 17.6 months (95% CI 13.8 to 23.0) in the osimertinib group and 9.6 months (95% CI 8.1 to 11.2) in the standard EGFR-TKI group.

New lesions

At the time of data cutoff, 62 patients (22%) on osimertinib had disease progression due to new lesions by investigator assessment compared to 114 patients (41%) receiving standard EGFR-TKI, mostly driven by CNS (5% vs.12%) and lung new lesions (7% vs.15%).

First sites of progression

The pattern of progression differed between the different types of lesions. Considering the higher number of progressions in the standard EGFR-TKI group, the relative contribution of progression was similar for target lesions (23% vs. 34%) and non-target lesions (14% vs. 25%), though there was a higher number of new lesions contributing to progression in the standard EGFR-TKI group (19% vs. 37%).

The proportional difference in the overall incidence of new lesions between the treatment groups was mostly driven by new lesions in the lung (osimertinib: 20 patients [7%]; standard EGFR-TKI: 41 [15%]); and the CNS (osimertinib: 15 [5%]; standard EGFR-TKI: 34 [12%]); new lesions in the liver occurred in 6 patients (2%) in the osimertinib group versus 20 patients (7%) in the standard EGFR-TKI group. New lesions in the bone were seen in a similar proportion of patients in both treatment groups (osimertinib: 11 [4%]; standard EGFR-TKI: 11 [4%]).

Adverse events (AEs)

Cardiac effects

In FLAURA, of those patients with a baseline and at least one follow-up LVEF assessment, LVEF decreased ≥10% and a drop to <50% occurred in 8 of 257 patients (3%) in the osimertinib group and 3 of 253 patients (1%) in the standard EGFR-TKI group.

Cardiac effects (cardiac failure) were reported in 12 patients (4%) in the osimertinib group and 6 patients (2%) in the standard EGFR-TKI group. Most adverse events in this category were ejection fraction decrease (10 of 12 patients in the osimertinib group vs. 5 of 6 patients in the standard EGFR-TKI group). This led to dose interruptions in 1 patient (<1%) in the osimertinib group and 2 patients (1%) in the standard EGFR-TKI group, with no cases of dose reduction. There was only one case of treatment discontinuation due to ejection fraction decrease, occurring in the standard EGFR-TKI group.

ECG QT prolongation made up the majority of AEs in the cardiac effects (QT) category (osimertinib group: 28 patients [10%]; standard EGFR-TKI group: 11 patients [4%]). No cases of arrhythmia were reported in the osimertinib group versus one case in the standard EGFR-TKI group. Dose interruptions and reductions due to QT prolongation were reported in 8 (3%) and 5 patients (2%) in the osimertinib group, respectively; in the standard EGFR-TKI group, dose interruptions and reductions were reported in 6 (2%) and 1 patients (<1%), respectively. Four patients (1%) in the osimertinib group and 1 patient (<1%) in the standard EGFR-TKI group discontinued treatment due to adverse events of ECG QT prolongation.

Interstitial lung disease (ILD)

Interstitial lung disease grouped terms were reported in 11 patients (4%) in the osimertinib group, including six cases of interstitial lung disease (2%) and five (2%) of pneumonitis. In the standard EGFR-TKI group, interstitial lung disease grouped terms were reported in 6 patients (2%), including 4 patients (1%) with interstitial lung disease and 2 (1%) with pneumonitis.

The median time to onset for ILD grouped-term adverse events was 106 days (range 9 to 425) in the osimertinib group and 83.5 days (range: 11 to 253) in the standard EGFR-TKI group. Patients mostly had an ILD outcome of resolving or recovered, with a similar proportion of patients reporting an outcome of recovered in both treatment groups (64% in the osimertinib group vs. 67% in the standard EGFR-TKI group). One patient (in the standard EGFR-TKI group) reported an outcome of not-recovered. No patients in either treatment group died from ILD or pneumonitis.

Adverse events leading to dose interruption, dose reduction or treatment discontinuation

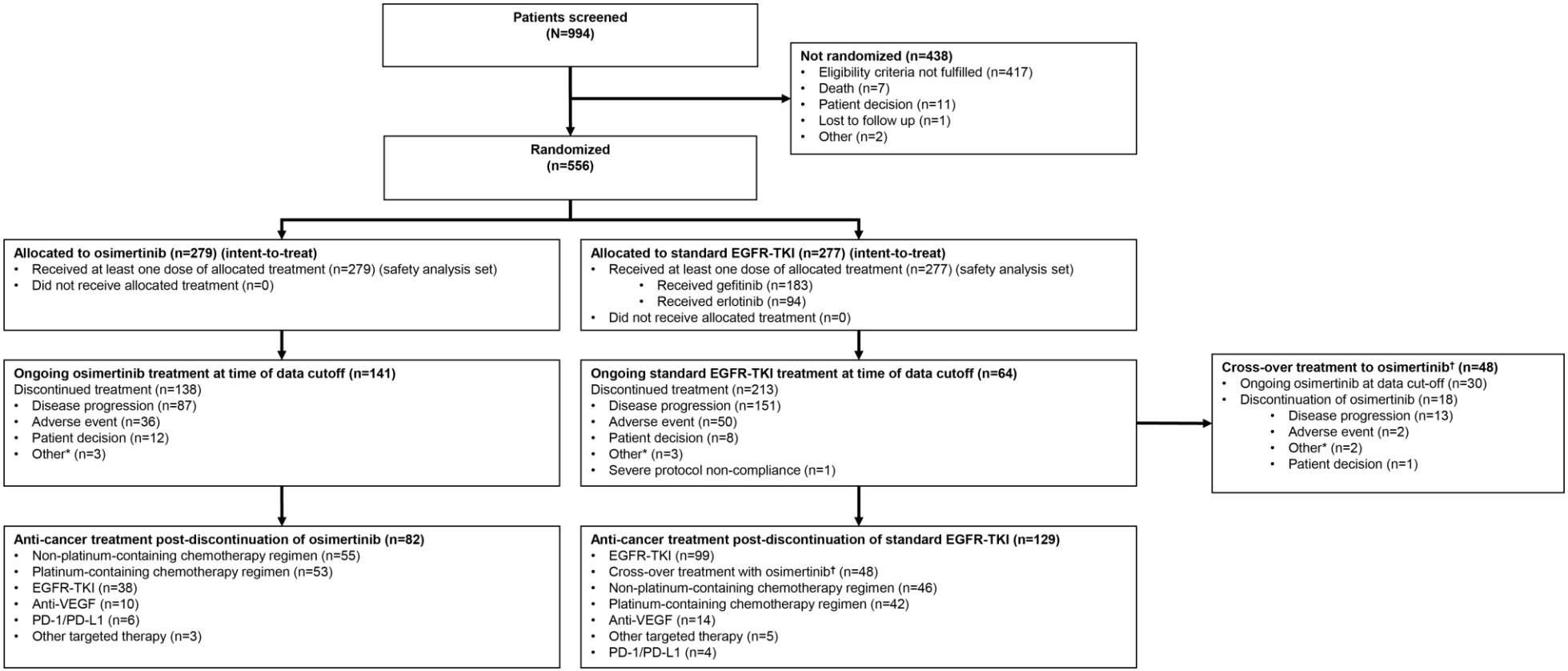
Overall, 70 patients (25%) in the osimertinib group and 66 patients (24%) in the standard EGFR-TKI group experienced AEs leading to dose interruptions. The most frequently experienced AEs leading to dose interruption in the osimertinib group were QT prolongation (8 patients), decreased appetite (7 patients), diarrhea (7 patients), and pneumonia (5 patients), while in the standard EGFR-TKI group dose interruption was driven by alanine aminotransferase increase (18 patients), aspartate aminotransferase increase (12 patients), QT prolongation (6 patients) and dermatitis acneiform (5 patients).

In total, 11 patients (4%) in the osimertinib group and 15 patients (5%) in the standard EGFR-TKI group experienced AEs leading to dose reduction. Dose reductions were driven mostly by QT prolongation (5 patients) and skin disorders (10 patients), in the osimertinib and standard EGFR-TKI groups respectively.

Adverse events leading to treatment discontinuation occurred at a lower frequency in the osimertinib group (37 patients [13%]) than in the standard EGFR-TKI group (49 patients [18%]). These were considered possibly causally-related by the investigator in a lower proportion of patients in the osimertinib group (27 patients [10%]) than the standard EGFR-TKI group (38 patients [14%]), largely due to a higher frequency of hepatic events in the standard EGFR-TKI group.

SUPPLEMENTARY FIGURES

Figure S1. Patient disposition



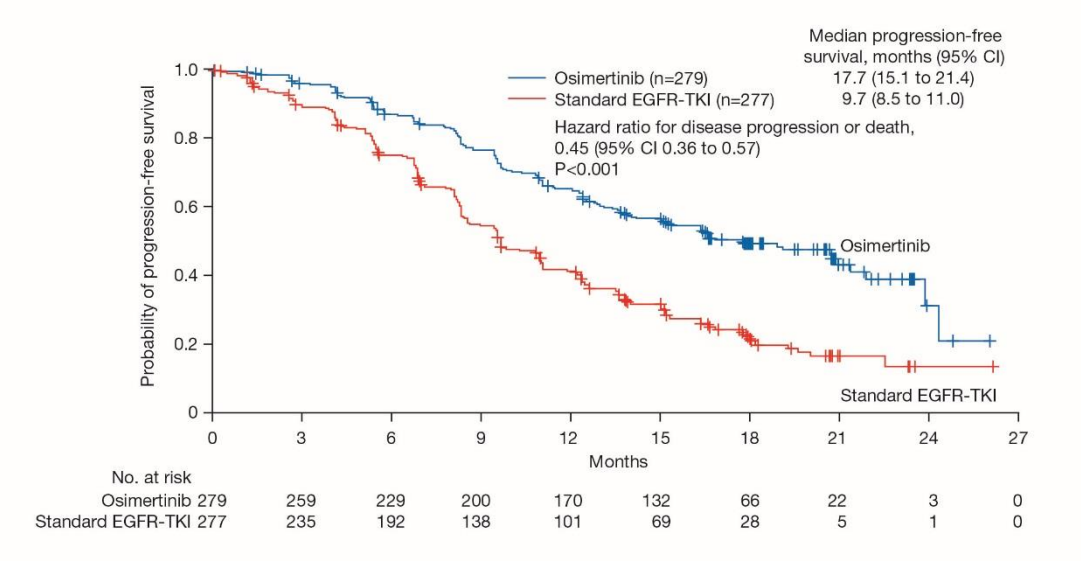
EGFR, epidermal growth factor receptor; PD1, programmed cell death; PD-L1, programmed death-ligand 1; TKI, tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor

* Any reason not specifically recorded; for example, subject died.

† Crossover patients are patients that crossed over and received at least one dose of open-label osimertinib.

Figure S2. Kaplan-Meier estimates of the duration of progression-free survival in the full analysis set, as assessed by blinded independent central review

Censored data are indicated by tick marks. For progression-free survival analysis, patients who had not progressed or died at the time of analysis were censored at the time of their last evaluable RECIST assessment.

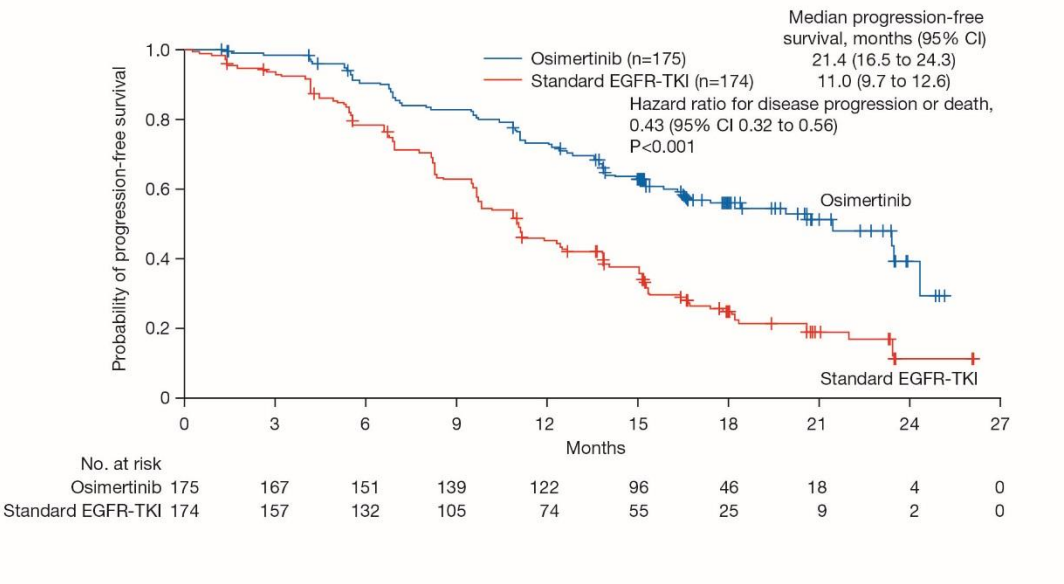


CI, confidence interval; RECIST, response evaluation criteria for solid tumors

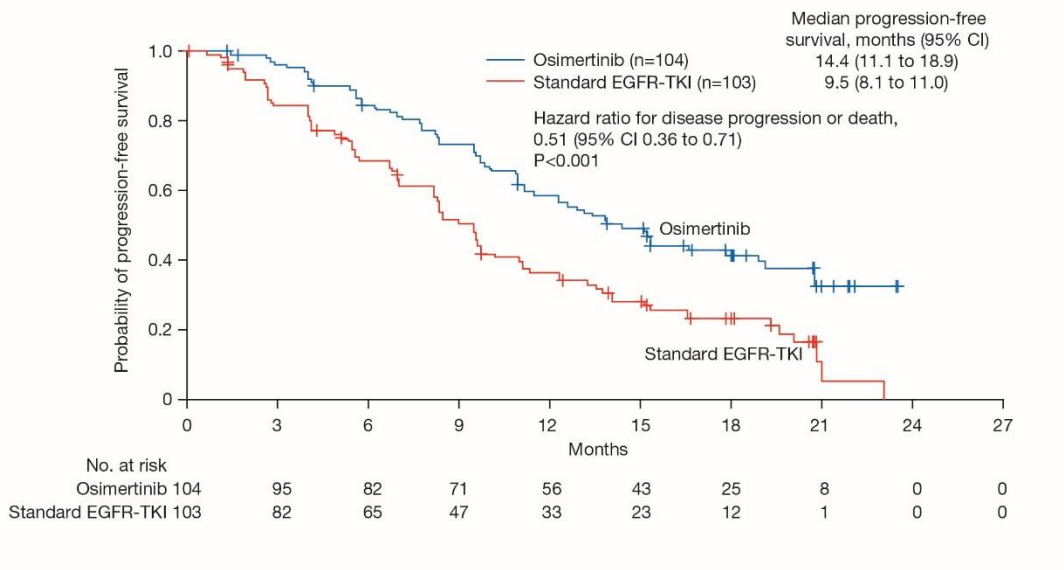
Figure S3. Duration of progression-free survival in the full analysis set, according to EGFR mutation

Shown are Kaplan–Meier estimates of the duration of progression-free survival as assessed by investigators in the full analysis set in patients with an *EGFR* exon 19 deletion mutation (Panel A), and in patients with an *EGFR* L858R mutation (Panel B). Censored data are indicated by tick marks. For progression-free survival analysis, patients who had survived and had no disease progression at the time of analysis were censored at the time of their last evaluable RECIST assessment.

A. Exon 19 deletion



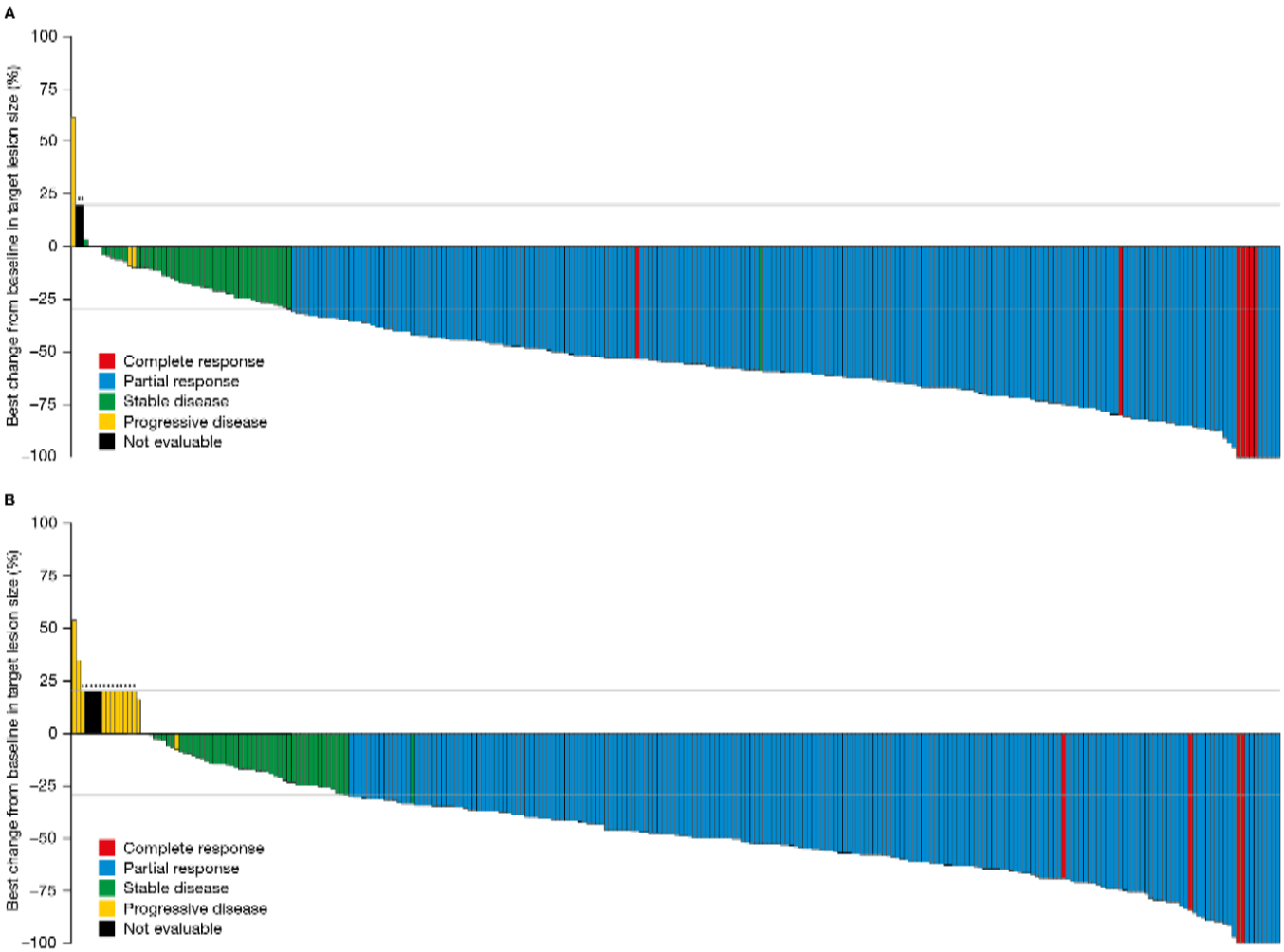
B. L858R



CI, confidence interval; RECIST, response evaluation criteria for solid tumors

Figure S4. Best percentage change from baseline in target lesions

Waterfall plots for best percentage change in target lesion size are shown for patients who received osimertinib (Panel A), versus patients who received standard EGFR-TKI (Panel B).



Best percentage change in target lesion size is the maximum reduction from baseline or the minimum increase from baseline in the absence of a reduction.

* Represents imputed values: If it is known that the patient has died, has new lesions or progression of non-target lesions, has withdrawn due to disease progression and has no evaluable target lesion (before or at progression) assessments, best change will be imputed as 20%.

SUPPLEMENTARY TABLES

Table S1. Summary of central EGFR mutation test results (screened population)

Result	Screened population (n=994)	Failed screening (n=438)	Randomized under central tissue result (n=289)	Randomized under local tissue result (n=267)
	Number (percent)			
EGFR mutation detected*	562 (57)	62 (14)	289 (100)	211 (79)
Exon 19 deletion	348 (35)	35 (8)	188 (65)	125 (47)
L858R	214 (22)	27 (6)	101 (35)	86 (32)
EGFR mutation not detected	230 (23)	224 (51)	0	6 (2)
Invalid	19 (2)	10 (2)	0	9 (3)
No sample†	183 (18)	142 (32)	0	41 (15)

EGFR, epidermal growth factor receptor
* Each patient is counted only once in the "EGFRm mutation detected" row.
† Patients in the no sample group were not tested due to: no tissue available, insufficient tissue, tissue failed pathology review, insufficient DNA yield.

Table S2. Patient disposition – crossover patients

Disposition	Number (%) of subjects
Patients who crossed over to osimertinib	**48**
Patients receiving ongoing osimertinib treatment at data cutoff	30 (63)
Patients who discontinued osimertinib	18 (38)
Patient decision	1 (2)
Adverse event	2 (4)
Severe non-compliance to protocol	0
Condition under investigation worsened	13 (27)
Other*	2 (4)
Patients ongoing trial at data cutoff	39 (81)
Patients who terminated trial	9 (19)
Subject decision	1 (2)
Death	8 (17)

* Any reason not specifically recorded; for example, subject died.
Crossover patients are patients that crossed over and received at least one dose of open-label osimertinib.

Table S3. First and second post-treatment anticancer therapy regimens (full analysis set)

Regimen	Osimertinib (n=279)	Standard EGFR-TKI (n=277)
First post-treatment anticancer therapy (including cross-over osimertinib), n (%)		
Discontinued randomized trial treatment	138 (49)	213 (77)
First post-treatment anticancer therapy	82 (29)	129 (47)
No post-treatment anticancer therapy	56 (20)	84 (30)
Ongoing randomized trial treatment	141 (51)	64 (23)
First post-treatment anticancer therapy, n (%)*		
EGFR-TKI	29 (21)	97 (46)
PD-1/PD-L1	3 (2)	2 (1)
Nonplatinum chemotherapy	50 (36)	27 (13)
Platinum-based chemotherapy	48 (35)	26 (12)
Other targeted therapy	2 (1)	3 (1)
Anti-VEGF	7 (5)	4 (2)
Second post-treatment anticancer therapy, n (%)		
Second post-treatment anticancer therapy	24 (9)	39 (14)
Only one post-treatment anticancer therapy	58 (21)	90 (32)
Second post-treatment anticancer therapy, n (%)*		
EGFR-TKI	10 (7)	14 (7)
PD-1/PD-L1	2 (1)	1 (<1)
Non-platinum chemotherapy	11 (8)	19 (9)
Platinum-based chemotherapy	4 (3)	16 (8)
Other targeted therapy	1 (1)	2 (1)
Anti-VEGF	3 (2)	6 (3)

EGFR, epidermal growth factor receptor; PD1, programmed cell death; PD-L1, programmed death-ligand 1; TKI, tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor

*The number of patients is shown with percentages calculated as the proportion of patients who discontinued randomized trial treatment.

The first post-treatment anticancer therapy was the first treatment started on or after the last dose date of randomized trial treatment. The second post-treatment anticancer therapy was the second treatment started on or after the last dose date of randomized trial treatment. Patients with no post-treatment anticancer therapy had discontinued randomized trial treatment without starting any other post-treatment anticancer therapy. Patients may be counted in multiple rows if they received more than one anticancer therapy or a combination therapy that contained drug substances from multiple classifications.

Table S4. Reason for CNS progression, by investigator assessment (full analysis set)

Reason for progression	Known/treated CNS metastases at trial entry		No known/treated CNS metastases at trial entry	
	Osimertinib (n=53)	Standard EGFR-TKI (n=63)	Osimertinib (n=226)	Standard EGFR-TKI (n=214)
	Number (percent)			
Total number of progression events	29 (55)	53 (84)	107 (47)	153 (71)
Number of patients with progression due to death*	4 (8)	4 (6)	7 (3)	10 (5)
Number of patients with CNS progression*	10 (19)	27 (43)	7 (3)	15 (7)
Progression in CNS only	3 (6)	10 (16)	4 (2)	12 (6)
New lesions only	3 (6)	9 (14)	4 (2)	12 (6)
Non-target lesions only	0	0	0	0
New and non-target lesions	0	1 (2)	0	0
Progression in CNS and non-CNS	7 (13)	17 (27)	3 (1)	3 (1)
New lesions only	0	0	1 (<1)	0
Non-target lesions only	4 (8)	6 (10)	0	0
New and non-target lesions	3 (6)	4 (6)	2 (1)	2 (1)
New and target lesions	0	2 (3)	0	1 (<1)
Target lesions and non-target lesions	0	1 (2)	0	0
New, target and non-target lesions	0	4 (6)	0	0
Number of patients with non-CNS progression only*	15 (28)	22 (35)	93 (41)	128 (60)

* Progression events that do not occur within two scheduled visits of the last evaluable assessment, or randomization, are censored and therefore excluded in the number of events (excluding deaths). All CNS lesions present at baseline are classified as non-target lesions, per protocol. Only non-CNS lesions can be classified as target lesions.

Table S5. Response to treatment by CNS metastases status

Response	Known/treated CNS metastases at trial entry		No known/treated CNS metastases at trial entry	
	Osimertinib (n=53)	**Standard EGFR-TKI (n=63)**	**Osimertinib (n=226)**	**Standard EGFR-TKI (n=214)**
Objective response rate, % (95% CI)	76 (62–86)	86 (75–93)	81 (75–86)	73 (66–79)
Odds ratio (95% CI)	0.5 (0.2–1.3) P=0.16		1.6 (1.0–2.5) P=0.04	
Type of response*, n (%)				
Complete response	0	1 (2)	7 (3)	3 (1)
Partial response	40 (75)	53 (84)	176 (78)	153 (71)
Stable disease ≥6 weeks	13 (25)	8 (13)	34 (15)	38 (18)
Progression	0	1 (2)	3 (1)	13 (6)
Not evaluable	0	0	6 (3)	7 (3)
	Osimertinib (n=40)	**Standard EGFR-TKI (n=54)**	**Osimertinib (n=183)**	**Standard EGFR-TKI (n=156)**
Duration of response, median months (95% CI)	13.8 (10.8–20.2)	8.3 (5.5–9.6)	17.6 (13.8–23.0)	9.6 (8.1–11.2)
Number of responders who subsequently progressed or died	22	46	84	112

CI, confidence interval; CNS, central nervous system; MTS, metastases; RECIST, response evaluation criteria in solid tumors

* Objective response rate is defined as the number (%) of patients with at least one visit response of complete response or partial response; confirmation of response was not required. Stable disease ≥6 weeks includes RECIST visit window (+/- 7 days). Statistical analysis was not performed for subgroup levels where there were <20 responses in both treatment groups or <5 events in one treatment group. The CIs were calculated using Clopper-Pearson exact method for binomial proportions. The analysis was performed using logistic regression with factors for treatment, CNS MTS, and a treatment by CNS MTS interaction. CI calculated using profile likelihood. An odds ratio >1 favours osimertinib. Duration of response was calculated as the time from the first documentation of complete response or partial response until the date of progression or death in the absence of RECIST progression, or the last evaluable RECIST assessment for patients that do not progress or die. Duration of response was calculated using the Kaplan-Meier technique.

Table S6. Grade ≥3 adverse events (safety analysis set)

Adverse event	Osimertinib n=279	Standard EGFR-TKI n=277
	Number (percent)	
Decreased appetite	7 (3)	5 (2)
Pneumonia	6 (2)	6 (2)
Diarrhea	6 (2)	7 (3)
Electrocardiogram QT prolonged	6 (2)	2 (1)
Hyponatremia	4 (1)	4 (1)
Pulmonary embolism	4 (1)	1 (<1)
Gamma-glutamyltransferase increased	4 (1)	0
Lymphocyte count decreased	4 (1)	0
Neutrophil count decreased	4 (1)	1 (<1)
Rashes and acnes†	3 (1)	19 (7)
Anemia	3 (1)	3 (1)
Hypokalemia	3 (1)	3 (1)
Interstitial lung disease†	3 (1)	4 (1)
Weight decreased	3 (1)	0
Urinary tract infection	2 (1)	0
Cataract	2 (1)	1 (<1)
Stomatitis	2 (1)	1 (<1)
Urticaria	2 (1)	0
Asthenia	2 (1)	2 (1)
Fatigue	2 (1)	2 (1)
Aspartate aminotransferase increased	2 (1)	12 (4)
Ejection fraction decreased	2 (1)	1 (<1)
Platelet count decreased	2 (1)	1 (<1)
Clostridial infection	1 (<1)	0
Gastoenteritis	1 (<1)	1 (<1)
Lower respiratory tract infection	1 (<1)	0
Paronychia†	1 (<1)	2 (1)
Pneumonia mycoplasmal	1 (<1)	0
Respiratory tract infection	1 (<1)	0
Skin infection	1 (<1)	0
Tracheobronchitis	1 (<1)	0
Urethritis	1 (<1)	0
Urinary tract infection bacterial	1 (<1)	1 (<1)
Endometrial cancer	1 (<1)	0
Histiocytic necrotising lymphadenitis	1 (<1)	0
Invasive ductal breast carcinoma	1 (<1)	0
Lymphopenia	1 (<1)	0
Neutropenia	1 (<1)	0
Pancytopenia	1 (<1)	0
Inappropriate antidiuretic hormone secretion	1 (<1)	0

Adverse event	Osimertinib n=279	Standard EGFR-TKI n=277
Diabetic ketoacidosis	1 (<1)	0
Hypermylasemia	1 (<1)	0
Hyperkalemia	1 (<1)	0
Hypermagnesemia	1 (<1)	0
Acute psychosis	1 (<1)	0
Anxiety	1 (<1)	0
Depression	1 (<1)	0
Cerebral infarction	1 (<1)	1 (<1)
Cognitive disorder	1 (<1)	1 (<1)
Depressed level of consciousness	1 (<1)	0
Hemorrhagic stroke	1 (<1)	0
Headache	1 (<1)	0
Transient ischemic attack	1 (<1)	0
Acute myocardial infarction	1 (<1)	0
Atrial fibrillation	1 (<1)	0
Cardiac arrest	1 (<1)	0
Cardiac failure chronic	1 (<1)	0
Myocardial infarction	1 (<1)	0
Deep vein thrombosis	1 (<1)	1 (<1)
Embolism	1 (<1)	0
Hypertension	1 (<1)	1 (<1)
Thrombophlebitis	1 (<1)	0
Dyspnea	1 (<1)	4 (1)
Hemoptysis	1 (<1)	1 (<1)
Hypoxia	1 (<1)	0
Pleural effusion	1 (<1)	1 (<1)
Abdominal pain	1 (<1)	0
Dysphagia	1 (<1)	0
Hemorrhoidal hemorrhage	1 (<1)	0
Intestinal ischemia	1 (<1)	0
Dry skin†	1 (<1)	3 (1)
Pruritus	1 (<1)	0
Acute kidney injury	1 (<1)	0
General physical health deterioration	1 (<1)	0
Alanine aminotransferase increased	1 (<1)	25 (9)
Transaminases increased	1 (<1)	0
White blood cell count decreased	1 (<1)	0
Head injury	1 (<1)	0
Subdural hematoma	1 (<1)	0
Anal abscess	0	1 (<1)
Bacteremia	0	1 (<1)
Bronchitis	0	1 (<1)
Clostridium difficile infection	0	1 (<1)
Cystitis	0	1 (<1)
Device-related infection	0	1 (<1)

Adverse event	Osimertinib n=279	Standard EGFR-TKI n=277
Endocarditis	0	1 (<1)
Lower respiratory tract infection viral	0	1 (<1)
Perineal abscess	0	1 (<1)
Pharyngotonsillitis	0	1 (<1)
Rectal abscess	0	1 (<1)
Sepsis	0	2 (1)
Sinusitis	0	1 (<1)
Endometrial adenocarcinoma	0	1 (<1)
Ovarian fibroma	0	1 (<1)
Prostate cancer	0	1 (<1)
Uterine leiomyoma	0	1 (<1)
Hyperglycemia	0	2 (1)
Hypoalbuminemia	0	2 (1)
Hypocalcemia	0	1 (<1)
Hypoglycemia	0	1 (<1)
Hypophosphatemia	0	1 (<1)
Restlessness	0	1 (<1)
Brain edema	0	1 (<1)
Cerebrovascular accident	0	1 (<1)
Migraine	0	1 (<1)
Parkinsonism	0	1 (<1)
Peripheral motor neuropathy	0	1 (<1)
Seizure	0	1 (<1)
Somnolence	0	1 (<1)
Blepharitis	0	1 (<1)
Vitreous detachment	0	1 (<1)
Vertigo	0	1 (<1)
Angina pectoris	0	1 (<1)
Bundle branch block left	0	1 (<1)
Pericardial effusion	0	1 (<1)
Ventricular extrasystoles	0	1 (<1)
Aortic dissection	0	1 (<1)
Circulatory collapse	0	1 (<1)
Peripheral artery occlusion	0	1 (<1)
Systolic hypertension	0	1 (<1)
Asthma	0	1 (<1)
Cough	0	1 (<1)
Pneumonia aspiration	0	1 (<1)
Pneumothorax	0	2 (1)
Pulmonary hemorrhage	0	1 (<1)
Respiratory failure	0	2 (1)
Gastrointestinal hemorrhage	0	1 (<1)
Melena	0	1 (<1)
Mouth ulceration	0	2 (1)
Vomiting	0	4 (1)

Adverse event	Osimertinib n=279	Standard EGFR-TKI n=277
Drug-induced liver injury	0	3 (1)
Hepatic failure	0	1 (<1)
Hepatic function abnormal	0	3 (1)
Henoch-Schonlein purpura	0	1 (<1)
Skin irritation	0	1 (<1)
Toxic epidermal necrolysis	0	1 (<1)
Musculoskeletal chest pain	0	1 (<1)
Pathologic fracture	0	2 (1)
Proteinuria	0	1 (<1)
Uterine polyp	0	1 (<1)
Condition aggravated	0	1 (<1)
Death	0	1 (<1)
Drug withdrawal syndrome	0	1 (<1)
Pyrexia	0	1 (<1)
Blood alkaline phosphatase increased	0	3 (1)
Blood bilirubin increased	0	1 (<1)
Protein urine present	0	1 (<1)
Femoral neck fracture	0	1 (<1)
Procedural pneumothorax	0	1 (<1)
Subarachnoid hemorrhage	0	1 (<1)

Safety analyses included all patients who received at least one dose of trial drug (safety analysis set). Includes adverse events with onset date on or after the date of first dose and up to and including 28 days following discontinuation of randomised treatment or the day before first administration of crossover treatment. †This category represents a grouped term for the event. If a patient had multiple preferred-term level events within a specific grouped term adverse event, then the maximum grade across those events was counted.

Table S7. Most common possibly causally-related adverse events (as assessed by the investigator) reported in at least 10% of patients treated with osimertinib or standard EGFR-TKI

Adverse events by preferred term*	Osimertinib (n=279)					Standard EGFR-TKI (n=277)				
	Any grade	Grade 1	Grade 2	Grade 3	Grade 4	Any grade	Grade 1	Grade 2	Grade 3	Grade 4
	Number (percent)									
Rashes and acnes†	152 (54)	125 (45)	24 (9)	3 (1)	0	205 (74)	105 (38)	81 (29)	19 (7)	0
Diarrhea	138 (49)	105 (38)	27 (10)	6 (2)	0	142 (51)	105 (38)	31 (11)	5 (2)	0
Dry Skin†	93 (33)	80 (29)	12 (4)	1 (<1)	0	92 (33)	70 (25)	19 (7)	3 (1)	0
Paronychia†	91 (33)	48 (17)	42 (15)	1 (<1)	0	84 (30)	52 (19)	30 (11)	2 (1)	0
Stomatitis	69 (25)	57 (20)	11 (4)	1 (<1)	0	45 (16)	36 (13)	8 (3)	1 (<1)	0
Decreased appetite	33 (12)	15 (5)	13 (5)	5 (2)	0	29 (10)	16 (6)	11 (4)	2 (1)	0
Pruritus	43 (15)	36 (13)	6 (2)	0	0	38 (14)	26 (9)	12 (4)	0	0
Aspartate aminotransferase elevation	22 (8)	15 (5)	5 (2)	2 (1)	0	57 (21)	31 (11)	16 (6)	10 (4)	0
Alanine aminotransferase elevation	17 (6)	11 (4)	5 (2)	1 (<1)	0	62 (22)	23 (8)	16 (6)	19 (7)	4 (1)

EGFR, epidermal growth factor receptor, TKI, tyrosine kinase inhibitor

* Adverse events occurring in 10% or more of patients in any group are listed. Safety analyses included all patients who received at least one dose of trial drug (safety analysis set). Some patients had more than one adverse event.

†This category represents a grouped term for the event. If a patient had multiple preferred-term level events within a specific grouped term adverse event, then the maximum grade across those events was counted.

Table S8. Serious adverse events (safety analysis set)

Serious adverse event	Osimertinib n=279	Standard EGFR-TKI n=277
	Number (percent)	
Pneumonia	7 (3)	7 (3)
Interstitial lung disease	4 (1)	3 (1)
Pulmonary embolism	4 (1)	1 (<1)
Pleural effusion	3 (1)	3 (1)
Gastroenteritis	2 (1)	1 (<1)
Decreased appetite	2 (1)	2 (1)
Deep vein thrombosis	2 (1)	0
Diarrhea	2 (1)	4 (1)
Enterocolitis	2 (1)	0
Asthenia	2 (1)	2 (1)
Pyrexia	2 (1)	2 (1)
Clostridial infection	1 (<1)	0
Clostridium difficile infection	1 (<1)	1 (<1)
Empyema	1 (<1)	0
Enteritis infectious	1 (<1)	0
Lower respiratory tract infection	1 (<1)	0
Pneumonia mycoplasmal	1 (<1)	0
Respiratory tract infection	1 (<1)	0
Sepsis	1 (<1)	3 (1)
Tracheobronchitis	1 (<1)	0
Urinary tract infection	1 (<1)	0
Histiocytic necrotising lymphadenitis	1 (<1)	0
Invasive ductal breast carcinoma	1 (<1)	0
Pancytopenia	1 (<1)	0
Inappropriate antidiuretic hormone secretion	1 (<1)	0
Diabetic ketoacidosis	1 (<1)	0
Acute psychosis	1 (<1)	0
Confusional state	1 (<1)	1 (<1)
Cerebral infarction	1 (<1)	1 (<1)
Cognitive disorder	1 (<1)	1 (<1)
Depressed level of consciousness	1 (<1)	0
Hemorrhagic stroke	1 (<1)	0
Spinal cord compression	1 (<1)	0
Transient ischemic attack	1 (<1)	0
Acute myocardial infarction	1 (<1)	0
Angina pectoris	1 (<1)	1 (<1)
Atrial fibrillation	1 (<1)	0
Cardiac arrest	1 (<1)	0
Cardiac failure chronic	1 (<1)	0
Cardiac tamponade	1 (<1)	0

Serious adverse event	Osimertinib n=279	Standard EGFR-TKI n=277
Myocardial infarction	1 (<1)	0
Tachyarrhythmia	1 (<1)	0
Embolism	1 (<1)	0
Thrombophlebitis	1 (<1)	0
Dyspnea	1 (<1)	3 (1)
Hypoxia	1 (<1)	0
Tonsillolith	1 (<1)	0
Abdominal pain	1 (<1)	0
Gastritis erosive	1 (<1)	0
Hemorrhoidal hemorrhage	1 (<1)	0
Inguinal hernia	1 (<1)	0
Intestinal ischemia	1 (<1)	0
Stomatitis	1 (<1)	0
Vomiting	1 (<1)	5 (2)
Back pain	1 (<1)	0
Rotator cuff syndrome	1 (<1)	0
Acute kidney injury	1 (<1)	0
Renal failure	1 (<1)	0
Alanine aminotransferase increased	1 (<1)	2 (1)
Aspartate aminotransferase increase	1 (<1)	1 (<1)
Electrocardiogram QT prolonged	1 (<1)	0
Platelet count decreased	1 (<1)	1 (<1)
Transaminases increased	1 (<1)	0
Femur fracture	1 (<1)	0
Head injury	1 (<1)	0
Subdural hematoma	1 (<1)	0
Wound complication	1 (<1)	0
Wrist fracture	1 (<1)	0
Anal abscess	0	1 (<1)
Bacteremia	0	1 (<1)
Bronchitis	0	1 (<1)
Cystitis	0	1 (<1)
Device-related infection	0	1 (<1)
Diarrhea infectious	0	1 (<1)
Diverticulitis	0	1 (<1)
Endocarditis	0	1 (<1)
Escherichia urinary tract infection	0	1 (<1)
Febrile infection	0	1 (<1)
Lower respiratory tract infection viral	0	1 (<1)
Perineal abscess	0	1 (<1)
Pharyngotonsillitis	0	1 (<1)
Pleural infection	0	1 (<1)
Rectal abscess	0	1 (<1)
Salmonella bacteremia	0	1 (<1)
Sinusitis	0	1 (<1)

Serious adverse event	Osimertinib n=279	Standard EGFR-TKI n=277
Endometrial adenocarcinoma	0	1 (<1)
Ovarian fibroma	0	1 (<1)
Prostate cancer	0	1 (<1)
Uterine leiomyoma	0	1 (<1)
Anemia	0	1 (<1)
Hyperglycemia	0	1 (<1)
Hypoglycemia	0	1 (<1)
Hyponatremia	0	1 (<1)
Brain edema	0	1 (<1)
Cerebrovascular accident	0	1 (<1)
Dizziness	0	1 (<1)
Ischemic stroke	0	1 (<1)
Migraine	0	1 (<1)
Peripheral motor neuropathy	0	1 (<1)
Seizure	0	1 (<1)
Vertigo	0	1 (<1)
Bundle branch block left	0	1 (<1)
Aortic dissection	0	1 (<1)
Circulatory collapse	0	1 (<1)
Orthostatic hypotension	0	1 (<1)
Peripheral artery occlusion	0	1 (<1)
Asthma	0	1 (<1)
Hemoptysis	0	1 (<1)
Pneumothorax	0	2 (1)
Pulmonary hemorrhage	0	1 (<1)
Respiratory failure	0	2 (1)
Chronic gastritis	0	1 (<1)
Gastrointestinal hemorrhage	0	1 (<1)
Melena	0	1 (<1)
Mouth ulceration	0	2 (1)
Drug-induced liver injury	0	3 (1)
Dermatitis acneiform	0	2 (1)
Toxic epidermal necrolysis	0	1 (<1)
Flank pain	0	1 (<1)
Musculoskeletal chest pain	0	1 (<1)
Pathologic fracture	0	1 (<1)
Ureterolithiasis	0	1 (<1)
Uterine polyp	0	1 (<1)
Chest pain	0	1 (<1)
Condition aggravated	0	1 (<1)
Death	0	1 (<1)
Drug withdrawal syndrome	0	1 (<1)
Fatigue	0	1 (<1)
General physical health deterioration	0	1 (<1)
Fall	0	1 (<1)

Serious adverse event	Osimertinib n=279	Standard EGFR-TKI n=277
Femoral neck fracture	0	1 (<1)
Procedural pneumothorax	0	1 (<1)
Subarachnoid hemorrhage	0	1 (<1)

Safety analyses included all patients who received at least one dose of trial drug (safety analysis set). Includes adverse events with onset date on or after the date of first dose and up to and including 28 days following discontinuation of randomized treatment or the day before first administration of crossover treatment.